

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester VI Examination, 2013-14

Paper No. BPT-601: Statistical Mechanics

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer all questions. Symbols have their usual meanings.***Question 1.** Answer any **four** of the following:

[4 × 5 = 20]

- (a) With reasons, find the number of distinct ways in which N objects, of which N_1 are indistinguishable of one type and N_2 are indistinguishable of a second type, can be accommodated in a total of $N = N_1 + N_2$ possible places.
- (b) The quantum distribution for an ideal Fermi gas is $f(\epsilon) = \frac{1}{e^{(\epsilon-\mu)/kT} + 1}$. What happens to this distribution in the limit $T \rightarrow 0$?
- (c) Consider two physical systems, which were separately in equilibrium, brought into thermal contact and allowed to interact. Formulate the condition for the two systems being in mutual thermodynamic equilibrium and hence derive the relation $S = k \ln \Omega$.
- (d) Considering a system in a grand canonical ensemble, show that the probability that the system be in a particle-energy (N_r, E_s) state would be:

$$P_{r,s} = \frac{\exp(-\alpha N_r - \beta E_s)}{\sum_{r,s} \exp(-\alpha N_r - \beta E_s)}$$

- (e) A two-energy-level system, with an energy gap of 0.1 eV, is in thermal equilibrium with a heat bath at 327°C. Obtain the probability that the system is in the higher energy level. (Given: $k = 8.617 \times 10^{-5}$ eV/K).
- (f) Consider a binomial distribution for a system for which $p = 1/2$, $q = 1/2$, $N = 6$. Determine the standard deviation σ and find the probability that n is in the range $\langle n \rangle - \sigma$ to $\langle n \rangle + \sigma$.

Question 2.

- (a) Consider a system in a canonical ensemble and show that the probability of the system being in energy state E_s is given by:

$$P_s = \frac{\exp(-\beta E_s)}{\sum_s \exp(-\beta E_s)}$$

[6.5]

- (b) Hence establish a relationship between the thermodynamics of the system and its partition function in the canonical ensemble.

[6]

OR

- (a) For an ideal monatomic classical gas, the partition function is given by:

$$Z_N = \frac{1}{N!h^{3N}}(2\pi mkT)^{3N/2}V^N$$

Show how this partition function resolves Gibbs' paradox. [6.5]

- (b) Show that the average energy of a classical monatomic gas is $\langle E \rangle = \frac{3}{2}NkT$. [6]

Question 3. Answer either (a) and (b) or (c) and (d):

- (a) Show that for a large number of random events, the binomial distribution reduces to the Gaussian distribution. Relate the characteristic parameters of the latter to those of the former. [6.5]
- (b) In a binomial distribution of total N steps out of which N_1 steps are to the right, show that the root-mean-square deviation is $\sigma = \sqrt{Npq}$. [6]

OR

- (a) What do you mean by phase space and density of states? Derive Liouville's theorem and discuss its consequences. [8]
- (b) Show that in the classical limit, quantum distribution laws, whether Fermi-Dirac or Bose-Einstein, reduce to the Maxwell-Boltzmann distribution. [4.5]

Question 4. Answer either (a) or (b):

- (a) Discuss the characteristic features of an ideal Fermi gas. Explain the meaning of Fermi energy. Show that the Fermi energy E_F varies as $n^{2/3}$ where $n = N/V$. Derive the relations for the Fermi gas ground state energy and ground state pressure. [12.5]

OR

- (a) Derive an expression for the partition function in Bose-Einstein statistics and hence obtain the relation for the mean number of particles $\langle n_s \rangle$ in a particular state s . Obtain the parameter α in terms of the chemical potential. Why is $\alpha = 0$ (or chemical potential $\mu = 0$) in the case of photons? [12.5]

Question 5.

- (a) Show that the relative root-mean-square fluctuation in energy E in a canonical ensemble varies as $1/\sqrt{N}$. What is the physical significance of this result? [4.5]
- (b) Discuss and compare the characteristic features of microcanonical, canonical, and grand canonical ensembles. [5]
- (c) Three gaseous systems composed of classical molecules, fermions, and bosons, respectively, are kept at 0 K. Which one of the systems will exert the greatest pressure? Explain. [3]

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